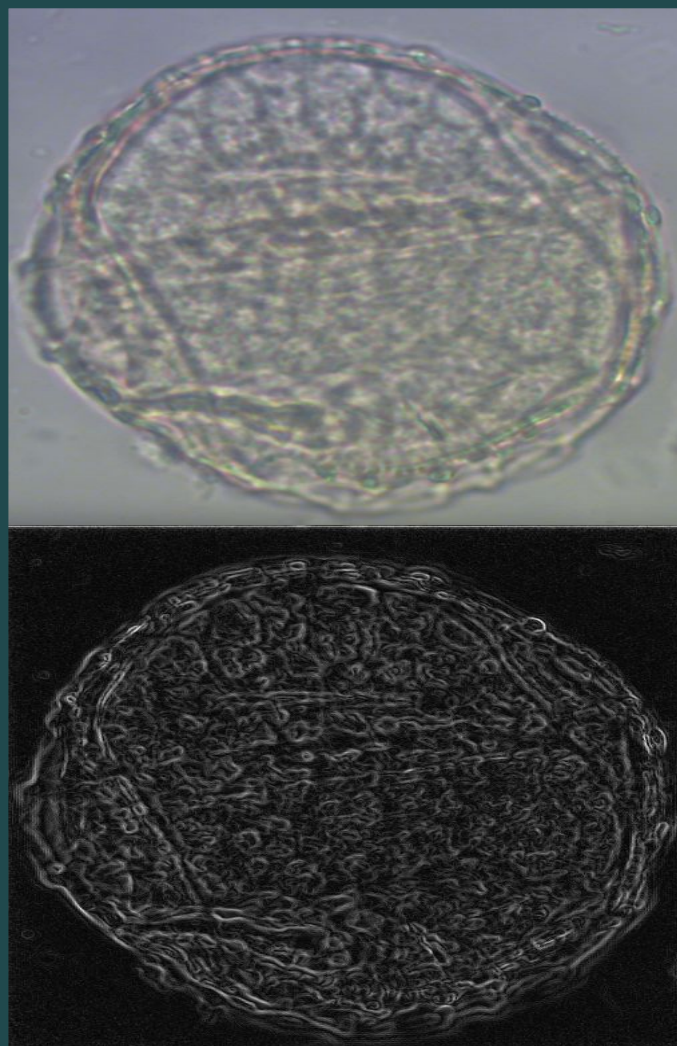


The Shape of Pollen: Using Computational Topology to Understand Plant Cells

By Jay Lebakken, Credit to Ben Croutteau



Content *overview*

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What is Palynology?

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Results/Conclusions

Background: What is Palynology?



[8]

[8]

Insights into paleoecology

The abundance of pollen in the fossil record makes it an amazing barometer for past ecosystems

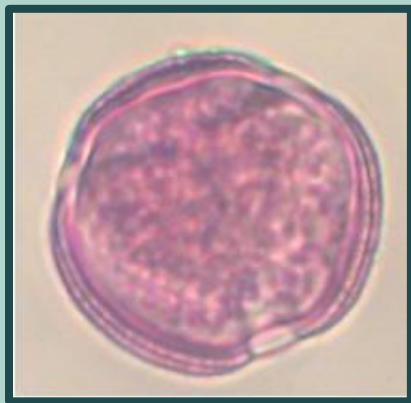
Applications in Climate Science

Past ecosystems give valuable insight into developing and understanding of contemporary plant life

Forensics

Pollen is everywhere, and pollen profiles can be used to assist in criminal and legal investigations by forensic scientists

[8]



THE PROBLEM

Pollen Classification is time consuming, and entirely manual

One slide of pollen can take up to *three days* of analysis by a single expert

[8]



THE PROBLEM

Pollen cells are squishy, 3d, and transparent

Because of this, pollen imagery can be highly variant, which complicates the task of classification

Current Research:

6

01

Convolutional Neural Networks

Classic approach in
computer vision for
classification.

02

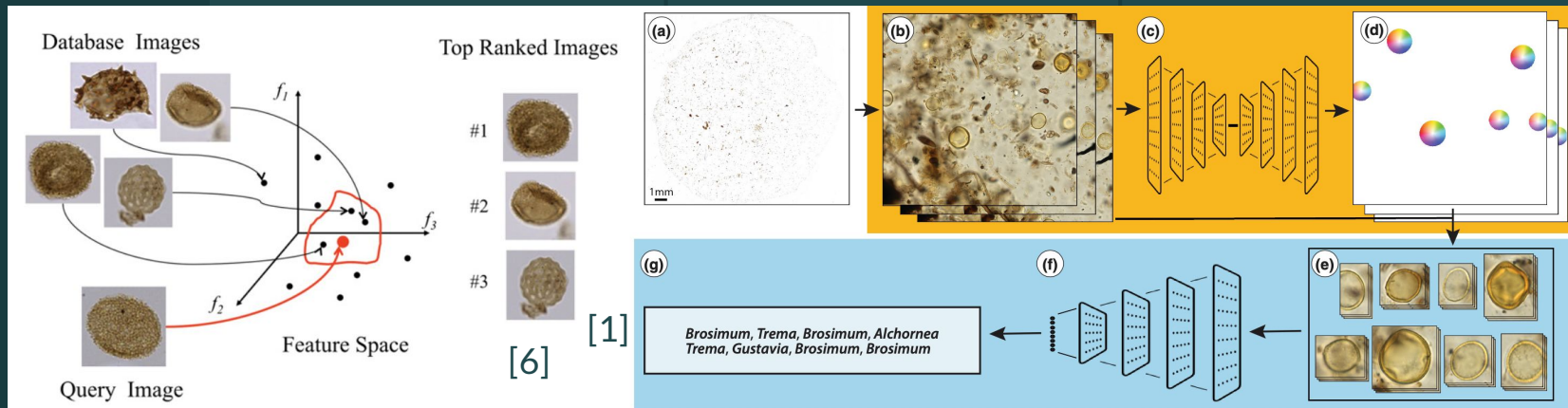
Subgraph Centrality

Textural algorithm on high
resolution images of pollen
surface

03

Manual Feature Space Creation

Manual tuning of feature
vectors based on insights
from experts



Our Approach: Topological Data Analysis

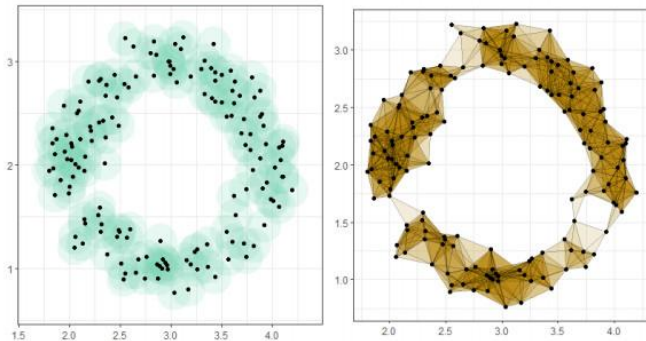


“Data has shape, shape has meaning, and meaning drives value”

Gunnar Carlsson

Stanford Professor, Math guy

The Idea: by looking at *holes* and *connected components*, we can extract meaning from pollen images



The Dataset

- Tropical (contemporary) grass pollen from Brazil
- Similar species
- Relatively low resolution images



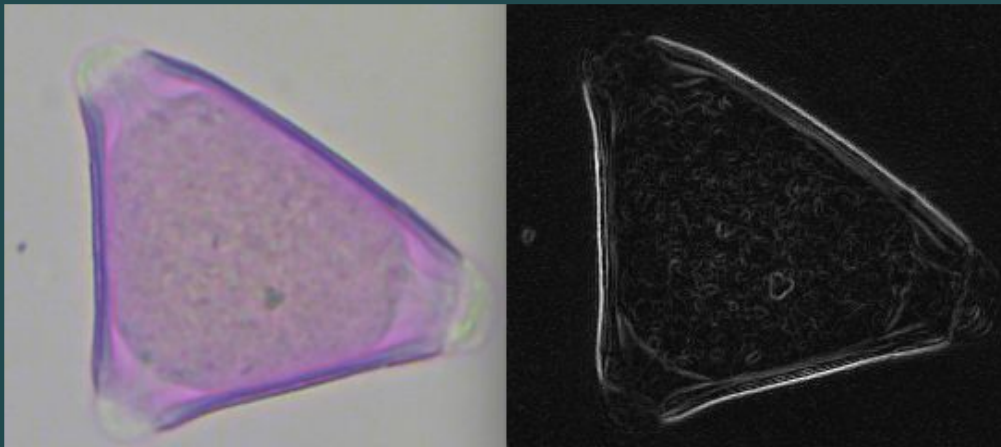
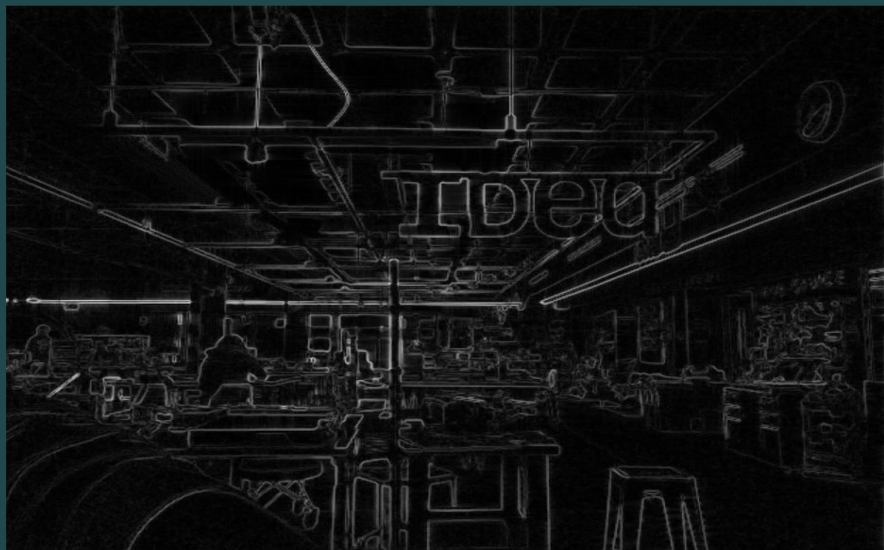
[8]

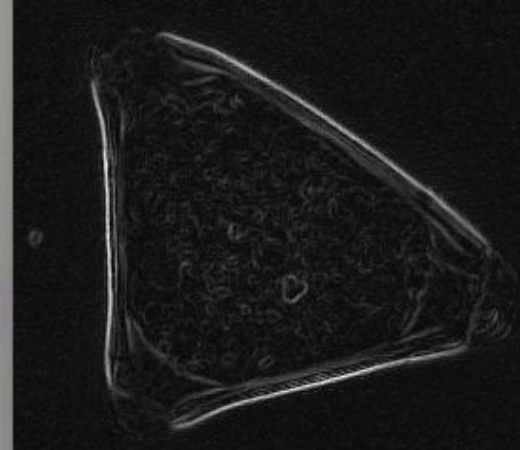
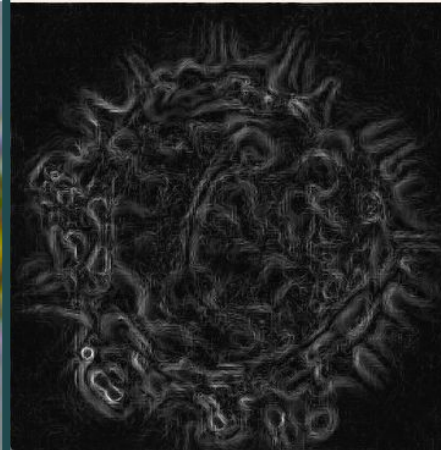
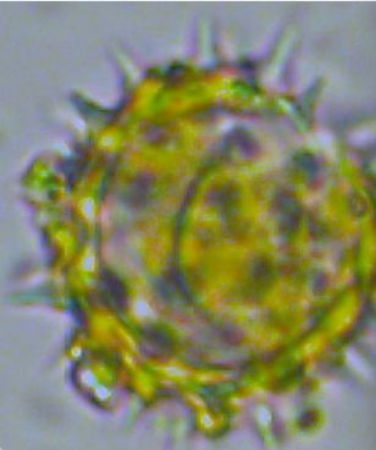


The Nature Conservancy: Cerrado

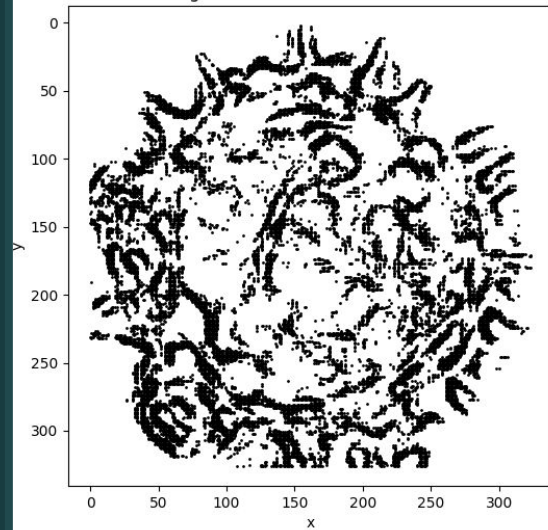
Data Pre-Processing

- Uses Fourier Transform to analyze the *frequency domain*
- Gradient is over the frequencies of the image

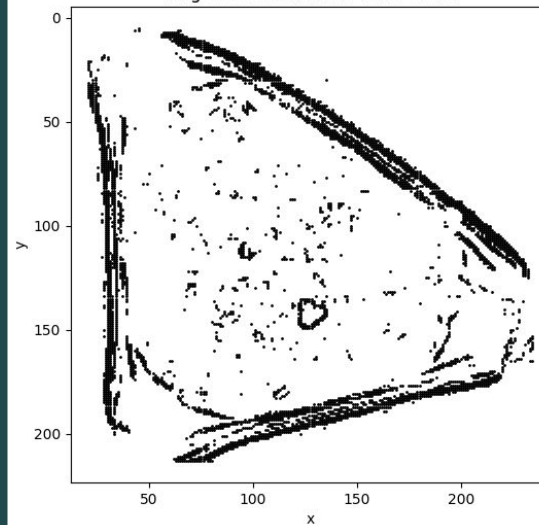


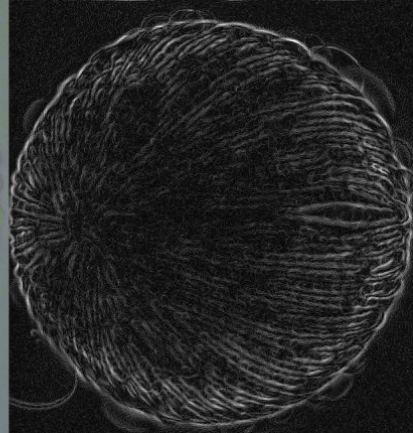
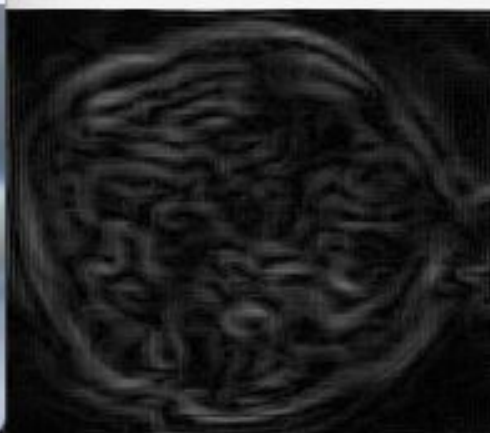
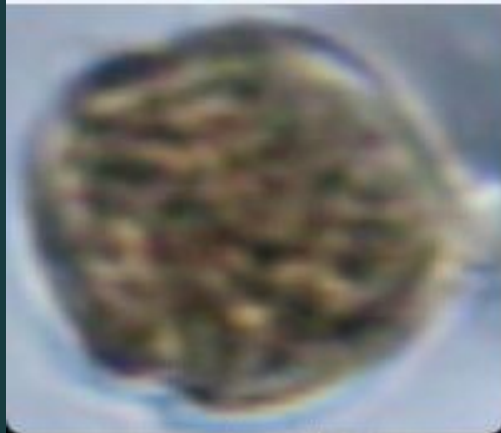


Edge Detection Pixels Point Cloud

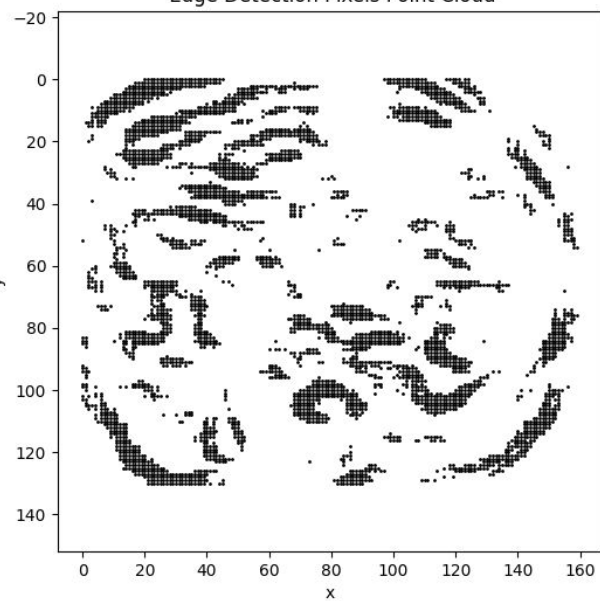


Edge Detection Pixels Point Cloud

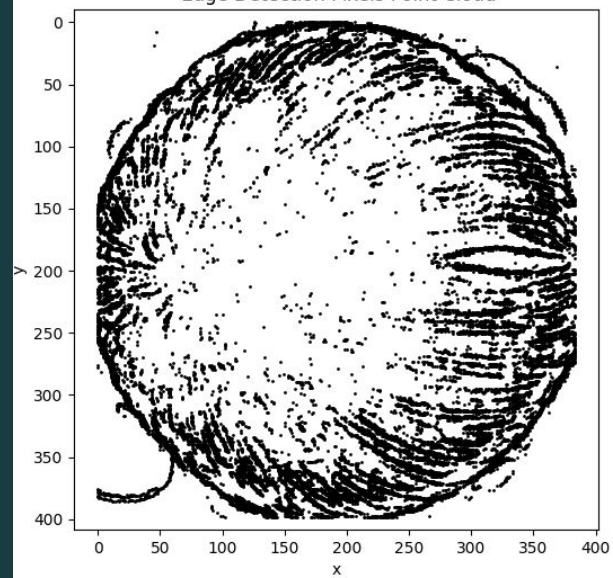


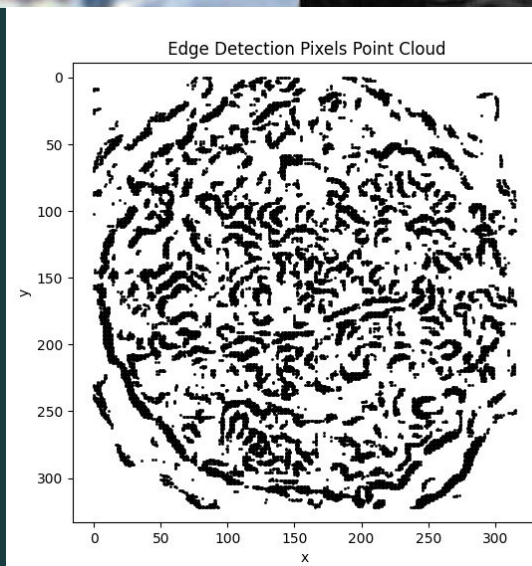
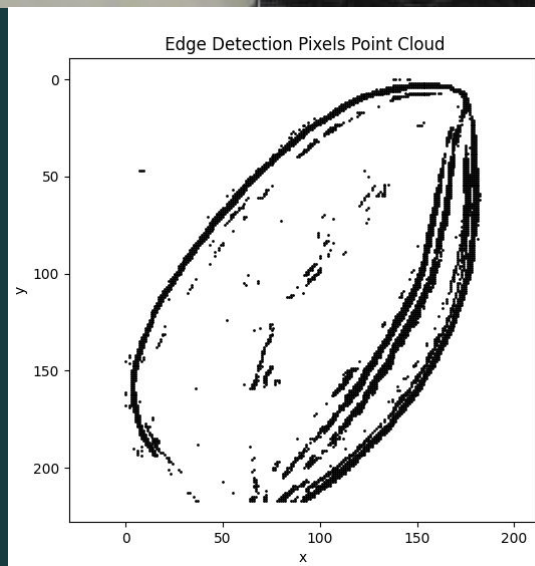
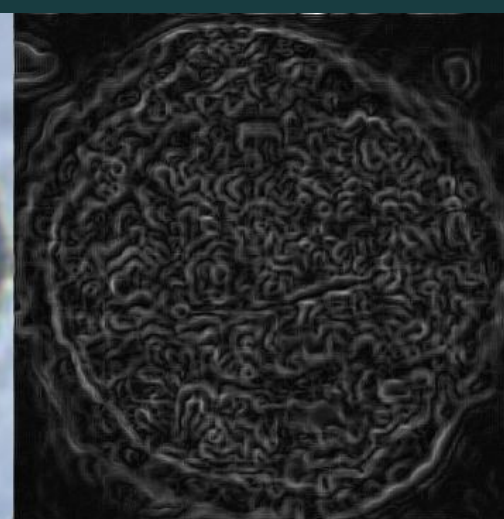


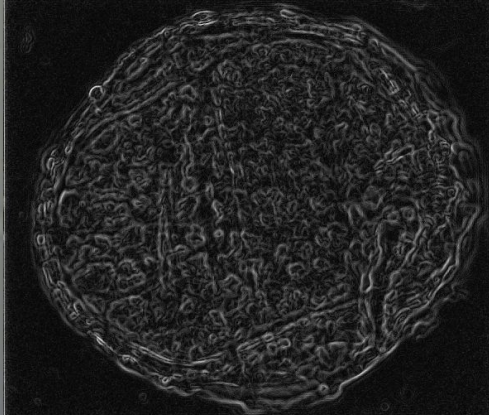
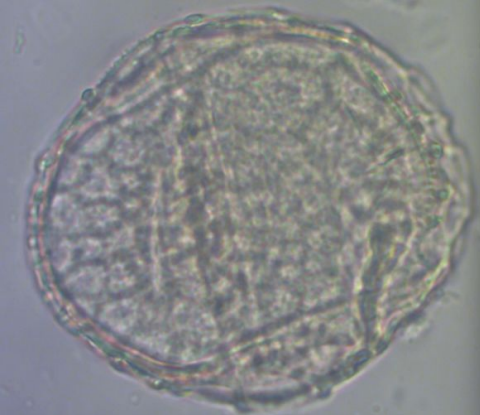
Edge Detection Pixels Point Cloud



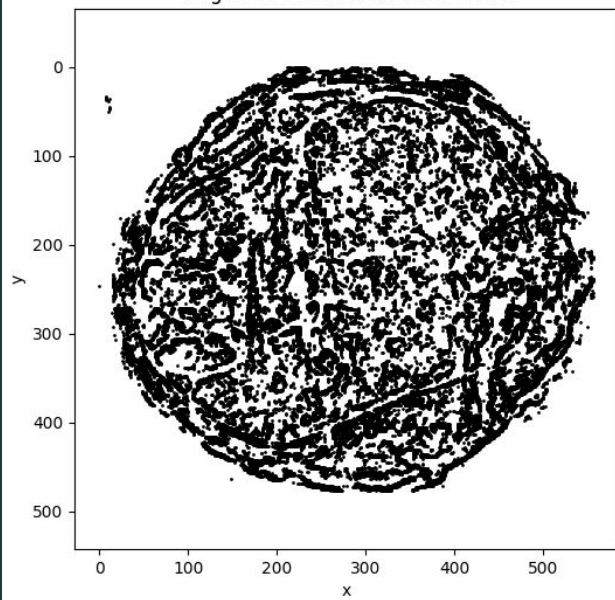
Edge Detection Pixels Point Cloud



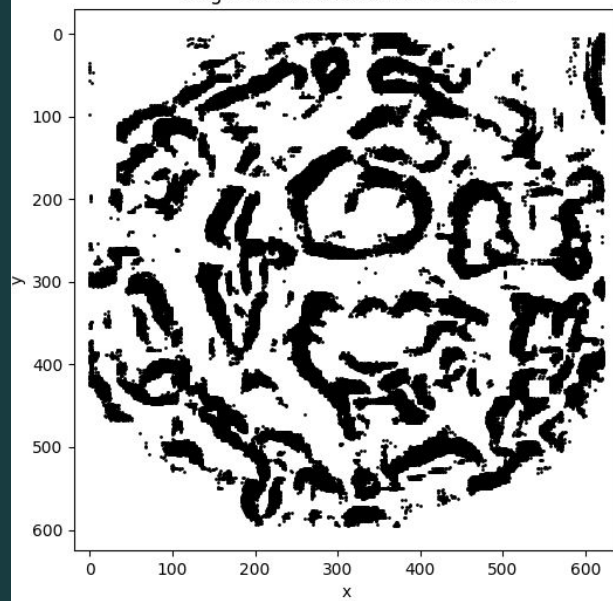




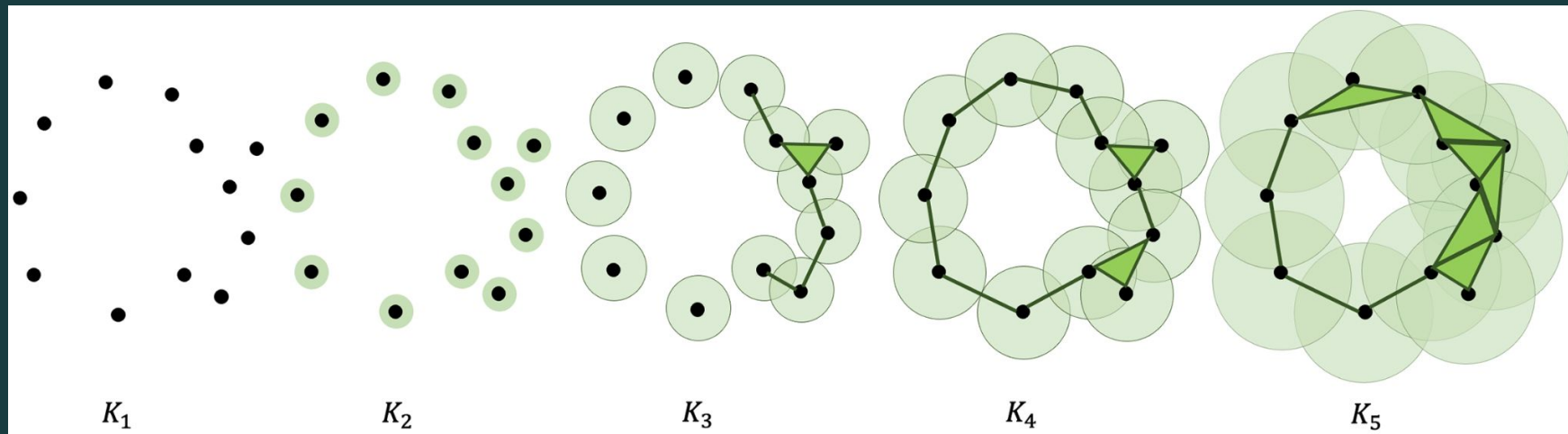
Edge Detection Pixels Point Cloud



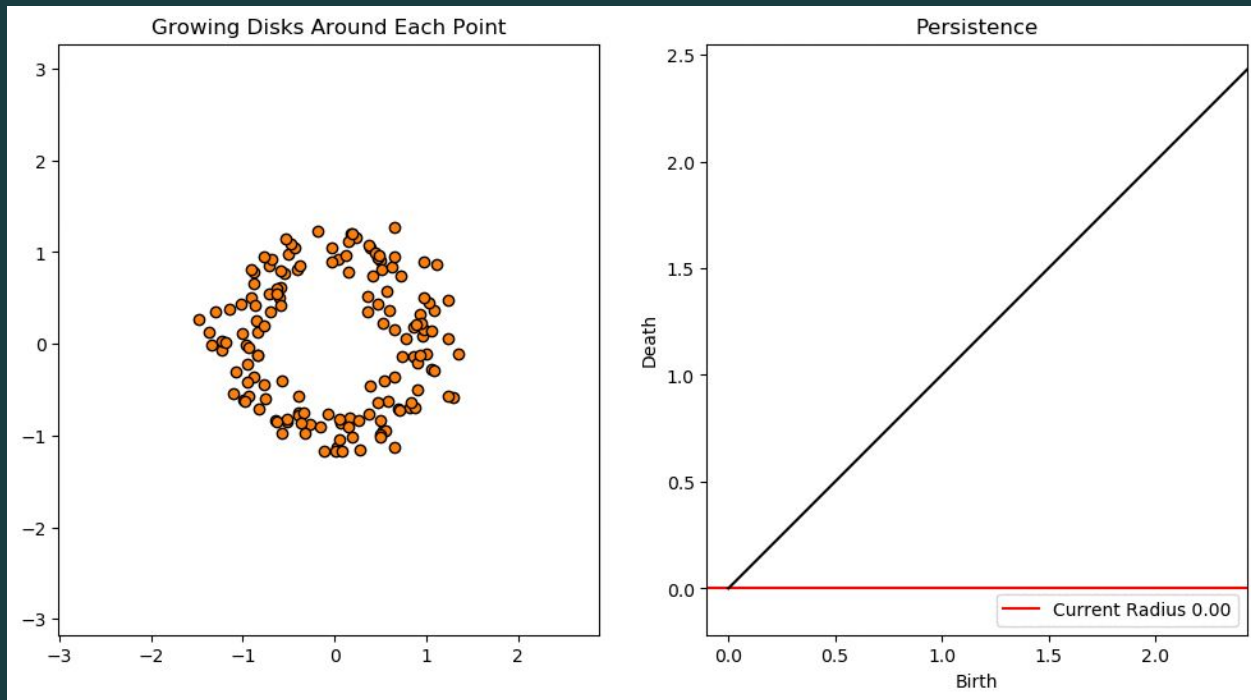
Edge Detection Pixels Point Cloud



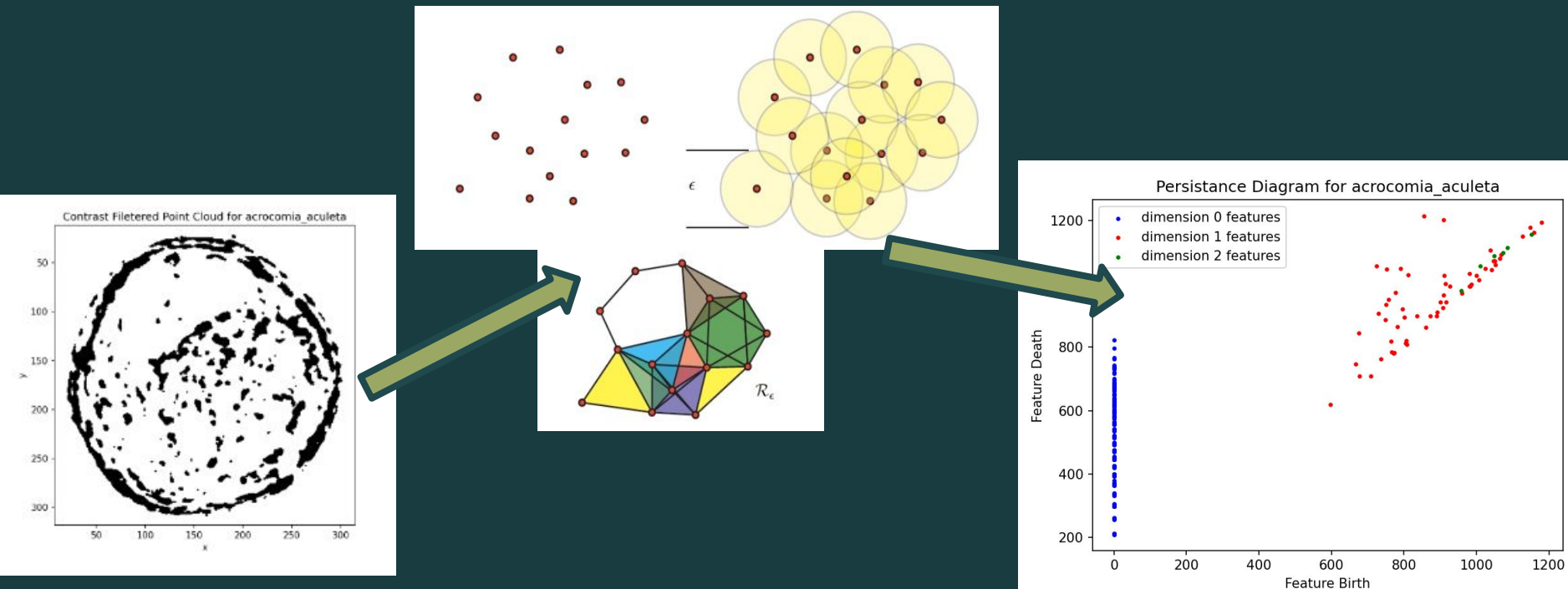
Persistent Homology



Persistent Homology



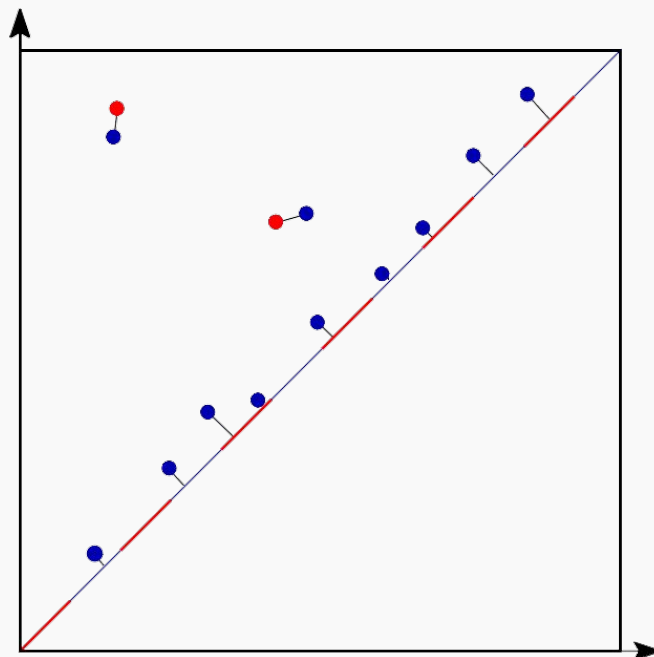
Data Processing Pipeline



Classification: Pairwise Bottleneck Distance

Upsides:

- Good measure of geometric distance between samples
- Simplistic
- More accurate than direct feature comparison



GUDHI: Bottleneck Distance

Downsides:

Insanely slow,
(roughly $O(n^4)$ for
each computation)

Results & Discussion

- Bottleneck distances between persistence diagrams classified samples on a limited (25%) dataset with an accuracy of 65.3%
- While this is a simplistic approach, it suggests that persistence diagrams are able to capture invariant features between pollen samples of the same species
- Unlike traditional approaches, these features do not have to be manually selected for a particular dataset
- Unfeasible for realistic use cases, but promising data exploration for future approaches

Applications:

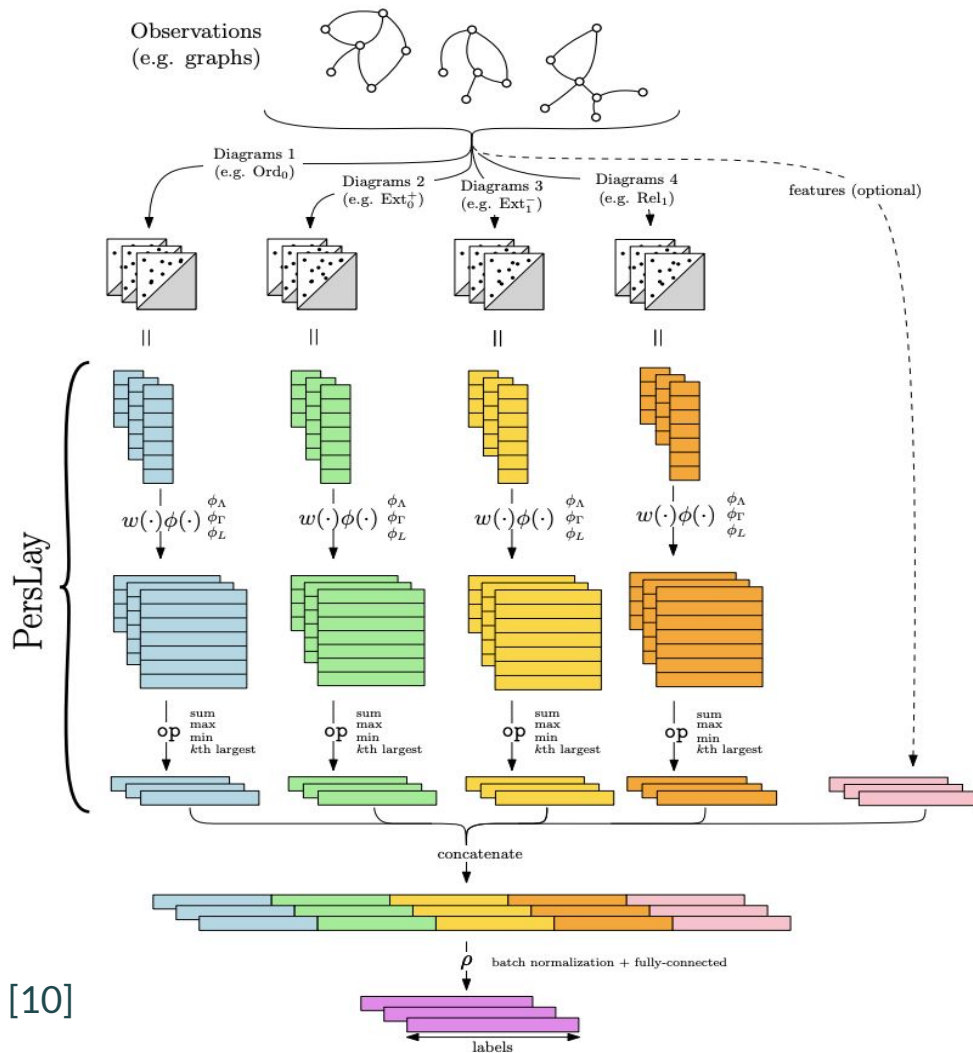
Classification: PerSlay Feature Selection

Upsides:

- quicker classification
- Captures information about persistence diagrams in vector form
- Strong Neural Network applications

Downsides:

- much harder to trace features
- more limited to the task of classification



$$\text{PERSLAY}(\text{Dg}) := \text{op}(\{w(p) \cdot \phi(p)\}_{p \in \text{Dg}}) \quad [10]$$

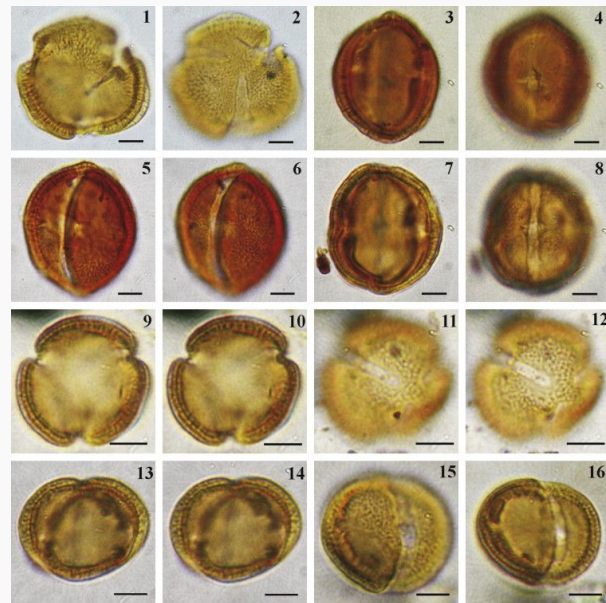
Conclusions

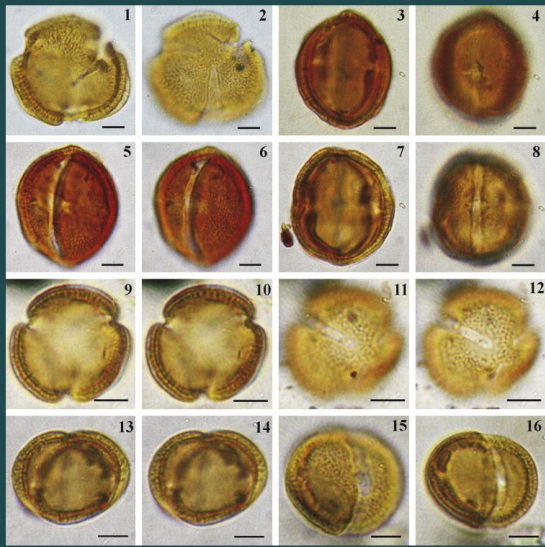
- Persistence Diagrams seem to capture essential information about pollen data even in low resolution settings
- Persistence Diagrams are a promising candidate for feature vectors of a neural network / unsupervised learning model
- Persistent homology is agnostic/resistant to many features that may trip up other classification models such as orientation, focus, and outline/shape

Future Directions

Persistence Diagrams capture information relevant to the task of classification:

- POLLENGEO
- ***The fossil question***
- Implement proposed model (perslay -> nn)





Thank you

Questions?